

BLDC Motors

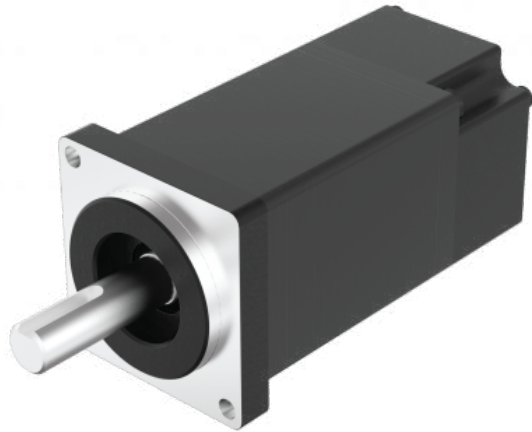


Parameter	Model 1	Model 2
Number of poles (—)	8	8
Number of phases (—)	3	3
Rated voltage (VDC)	24	24
Rated torque (mN·m)	8	20
Continuous stall torque (mN·m)	9.6	24
Rated speed (RPM)	4600	3800
Rated power (W)	3.8	8
Peak torque (mN·m)	24	60
Peak current (A)	1.1	2
Line-to-line resistance (Ohms @ 20°C)	23	11.6
Line-to-line inductance (mH)	6.2	4.3
Torque constant (mN·m/A)	29.7	35.8
Back EMF (Vrms/KRPM)	2.2	2.65
Rotor inertia (g·cm²)	0.66	1.32
Motor length – L (mm)	45	68
Weight (kg)	0.07	0.12

Parameter	Model 1	Model 2	Model 3
Number of poles (—)	4	4	4
Number of phases (—)	3	3	3
Rated voltage (VDC)	15	24	24
Rated torque (mN·m)	5	14	50
Continuous stall torque (mN·m)	6	17	60
Rated speed (RPM)	8000	10000	4000
Rated power (W)	4	15	21
Peak torque (mN·m)	15	42	150
Peak current (A)	1.1	2.5	4.1
Line-to-line resistance (Ohms @ 20°C)	8.2	4.2	4.2
Line-to-line inductance (mH)	2.3	1.69	2.3
Torque constant (mN·m/A)	14.3	16.9	37.4
Back EMF (Vrms/KRPM)	1.06	1.25	2.77
Rotor inertia (g·cm²)	2.35	3.69	10.98
Motor length – L (mm)	28	38	77
Weight (kg)	0.06	0.082	0.28

Parameter	Model 1
Number of poles (—)	14 / 14
Number of phases (—)	3
Rated voltage (VDC)	12 / 24
Rated torque (mN·m)	43.3 / 52.8
Continuous stall torque (mN·m)	52 / 63
Rated speed (RPM)	9660 / 10300
Rated power (W)	44 / 57
Peak torque (mN·m)	120 / 158
Peak current (A)	13.4 / 9.35
Line-to-line resistance (Ohms @20°C)	0.227 / 0.501
Line-to-line inductance (mH)	0.109 / 0.39
Torque constant (Nm/A)	8.95 / 16.9
Back EMF (Vrms/KRPM)	0.66 / 1.25
Rotor inertia (g·cm²)	10.5 / 10.5
Motor length –L (mm)	26 / 26
Weight (kg)	0.17 / 0.17

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Parameter	Models
Rated power (W)	50 / 100
Rated voltage (VAC/VDC)	220 / 48
Rated speed (rpm)	3000 / 3000
Rated torque (N·m)	0.16 / 0.16
Rated current (Arms)	0.2 / 1.3
Peak torque (N·m)	0.48 / 0.48
Peak current (Arms)	0.6 / 3.9
Line-to-line resistance (Ohm)	248 / 5.8
Line-to-line inductance (mH)	160 / 3.8
Back EMF constant (V/KRPM)	54.5 / 8.5
Torque constant (N·m/A)	0.80 / 0.123
Rotor inertia (kg·cm ²)	0.024 / 0.024
Rotor inertia with brake (kg·cm ²)	0.034 / 0.034
Number of poles (—)	10
Motor length L1 (mm)	72 / 88
Motor length with brake L2 (mm)	106 / 122
Weight (kg)	0.4 / 0.5
Weight with brake (kg)	0.65/ 0.75

Parameter	Models
Rated power (W)	200 / 400
Rated voltage (VDC)	48 / 220
Rated speed (RPM)	3000 / 3000
Rated torque (N·m)	0.64 / 1.27
Rated current (A)	6 / 1.2
Peak torque (N·m)	1.92 / 3.81
Peak current (A)	18 / 3.6
Line-to-line resistance (Ω)	0.38 / 9.35
Line-to-line inductance (mH)	0.68 / 17
Back EMF (V/KRPM)	6.5 / 32
Torque constant (Nm/A)	0.107 / 0.53
Rotor inertia Jm0 (kg·cm ²)	0.15 / 0.25
Rotor inertia with brake Jm1 (kg·cm ²)	0.15 / 0.25
Number of poles (—)	8
Motor length L1 (mm)	86.5 / 79.5
Motor length with brake L2 (mm)	123 / 116
Weight (kg)	0.95 / 0.9
Weight with brake (kg)	1.35 / 1.3

Parameter	Models
Number of poles (—)	10
Number of phases (—)	3
Rated voltage (VDC)	310 / 48
Rated torque (N·m)	2.39 / 2.39
Continuous stall torque (N·m)	2.86 / 2.86
Rated speed (RPM)	3000 / 3000
Rated power (W)	750 / 1000
Peak torque (N·m)	7.14 / 9.6
Peak current (A)	13.2 / 73.5
Line-to-line resistance (Ω @20°C)	4.09 / 0.089
Line-to-line inductance (mH)	12.6 / 0.275
Torque constant (N·m/A)	0.736 / 0.115
Back EMF (Vrms/KRPM)	54.5 / 8.5
Rotor inertia (kg·cm ²)	1.56 / 2.22
Motor length L (mm)	112 / 118
Weight (kg)	1.56 / 2.22

Parameter	Models
Rated Power (W)	2000 / 3000
Rated Voltage (VAC/VDC)	220 / 48
Rated Speed (RPM)	2000 / 2000
Rated Torque (N·m)	9.55 / 14.3
Rated Current (Arms)	7.8 / 52.9
Peak Torque (N·m)	28.65 / 42.9
Peak Current (Arms)	23.4 / 158.7
Line-to-Line Resistance (Ω)	1.4 / 0.03
Line-to-Line Inductance (mH)	6.8 / 0.14
Back EMF Constant (V/KRPM)	81 / 12
Torque Constant (N·m/A)	1.22 / 0.18
Rotor Inertia (kg·cm ²)	14 / 19.9
Rotor Inertia with Brake (kg·cm ²)	15.2 / 20.2
Number of poles (—)	10
Motor Length L1 (mm)	150 / 165
Motor Length with Brake L2 (mm)	178 / 193
Weight (kg)	7 / 11.7
Weight with Brake (kg)	9.5 / 14.2

The motor can be designed &manufactured with customized request.